

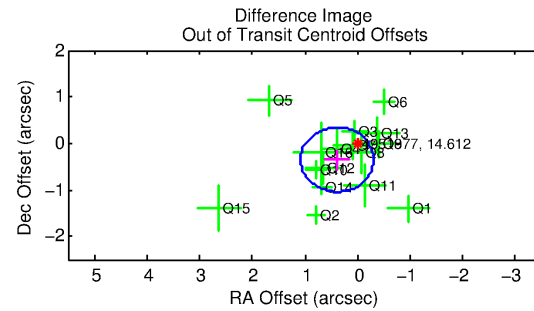
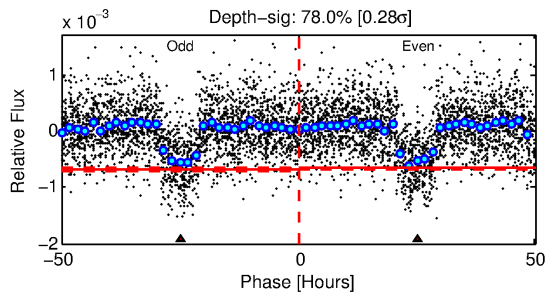
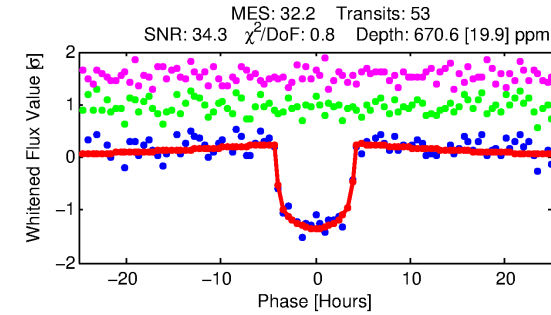
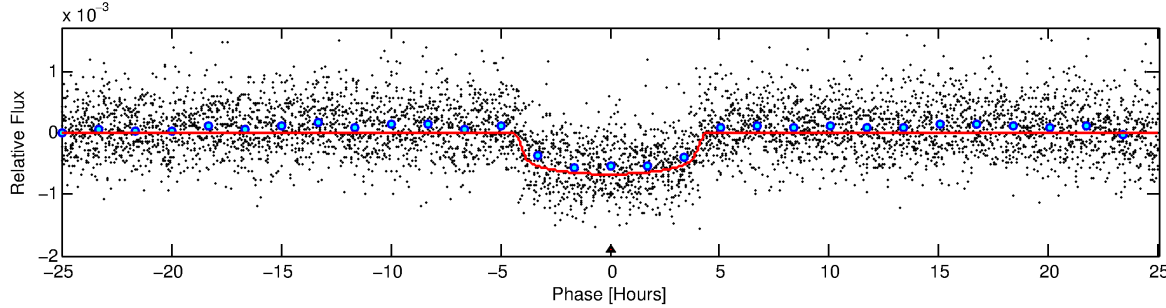
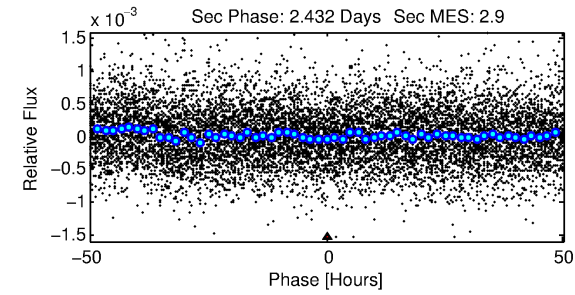
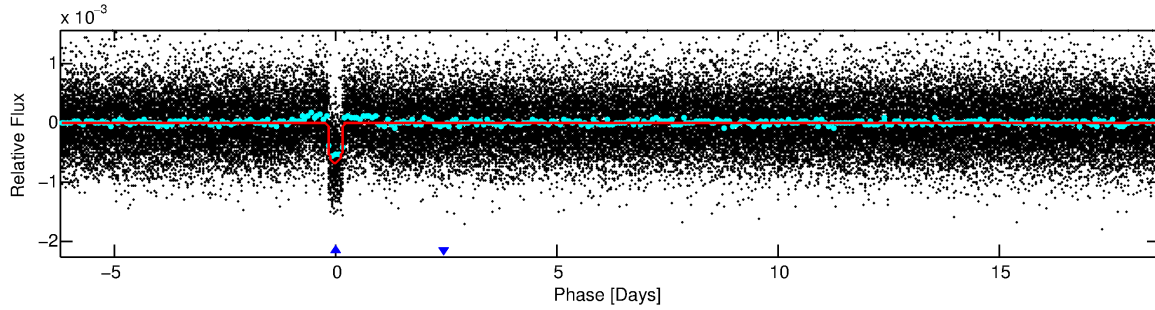
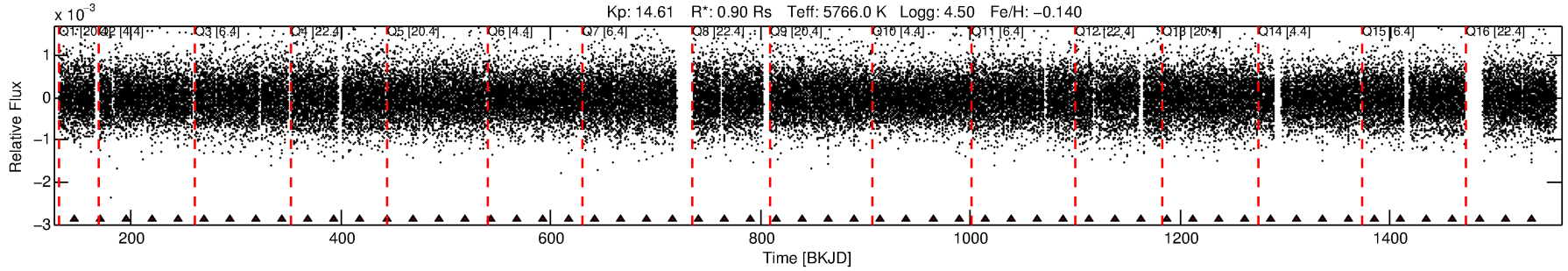
KIC 004951877

Link to DV Report: http://exoplanetarchive.ipac.caltech.edu/data/KeplerData/004/004951/004951877/dv/kplr004951877-20130815195353_dvr.pdf

TCE	KOI?	Period (Days)	T ₀ (BKJD)	Depth (ppm)	Duration (Hours)	MES
004951877-01	Yes	24.7963	145.528	670.6	8.3467	32.2

Flux Vetting: DV One-Page Summary

KIC: 4951877 Candidate: 1 of 1 Period: 24.796 d



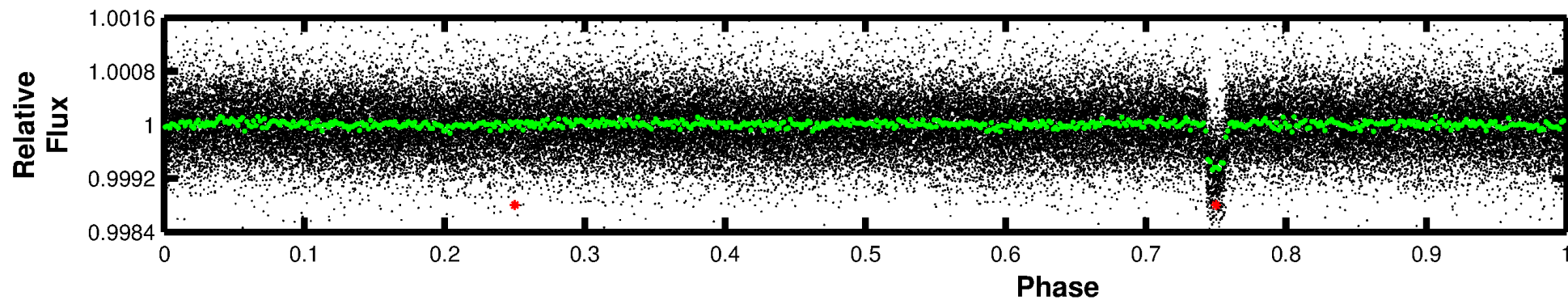
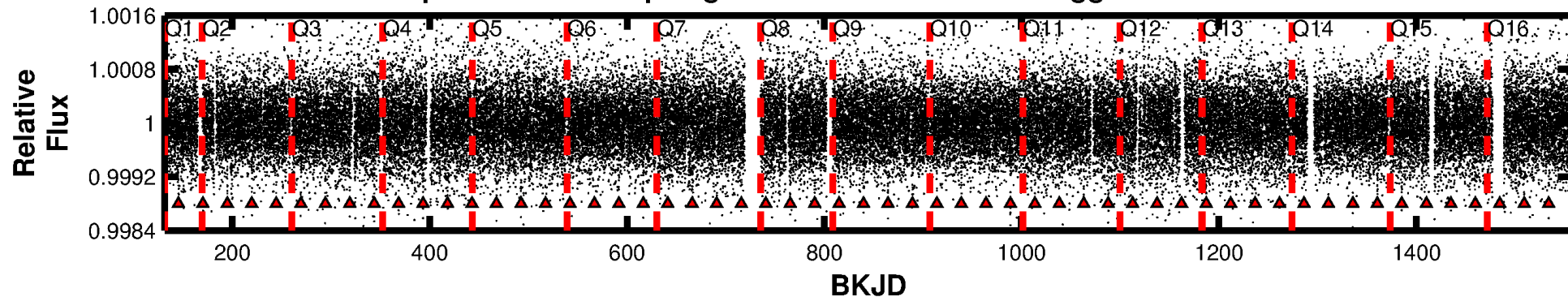
DV Fit Results:

Period = 24.79629 [0.00011] d
Epoch = 145.5279 [0.0037] BKJD
 $R_p/R^* = 0.0235$ [0.0131]
 $a/R^* = 23.10$ [57.45]
 $b = 0.11$ [22.77]
 $T_{eq} = 598$ K
 $R_p = 2.31$ Re
 $a = 0.1627$ AU

DV Diagnostic Results:

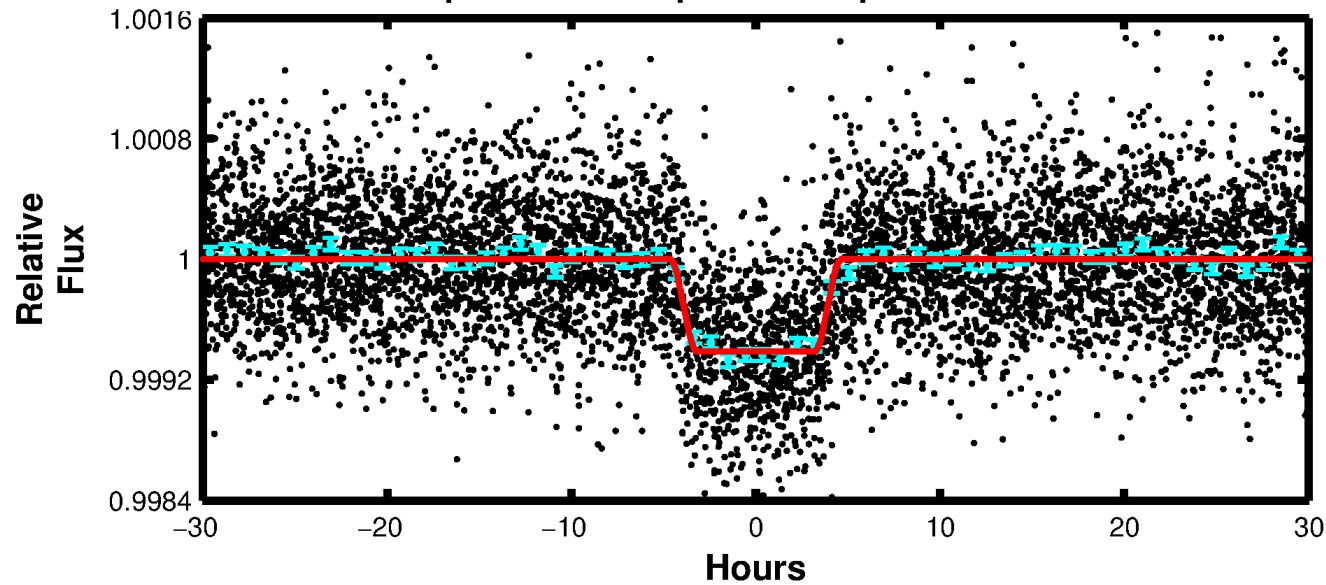
Epoch-sig: 99.5% [0.01σ]
ShortPeriod-sig: N/A
LongPeriod-sig: N/A
Bootstrap-pfa: N/A
Centroid-sig: 0.0%
Centroid-so: 0.138 arcsec [0.42σ]
OotOffset-rm: 0.525 arcsec [2.28σ]
KicOffset-rm: 0.576 arcsec [2.53σ]
OotOffset-bf: N/A
KicOffset-bf: N/A
OotOffset-st: 4/4/4/4 [16]
KicOffset-st: 4/4/4/4 [16]
DiffImageQuality-fgm: 1.00 [16/16 of 16]

KepId: 4951877 Kpmag: 14.61 Teff: 5766.0 Logg: 4.50 Rstar: 0.90



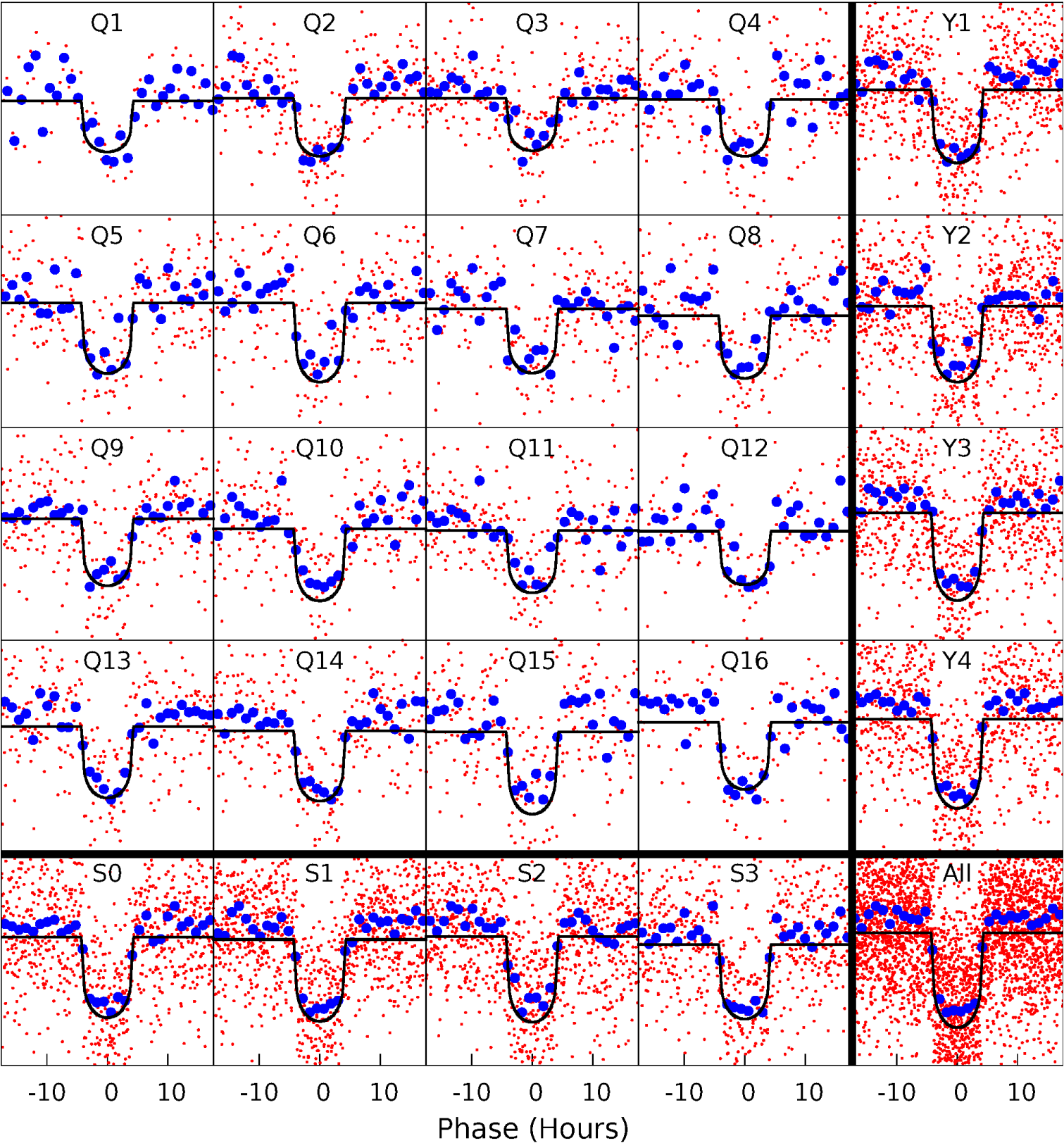
Flux Vetting - Alternate Detrending with Trapezoid Fit

Per: 24.797 Dep: 648.7 Teq: 598.1 Rp: 2.50 b: -1.00 SNR: 36.91



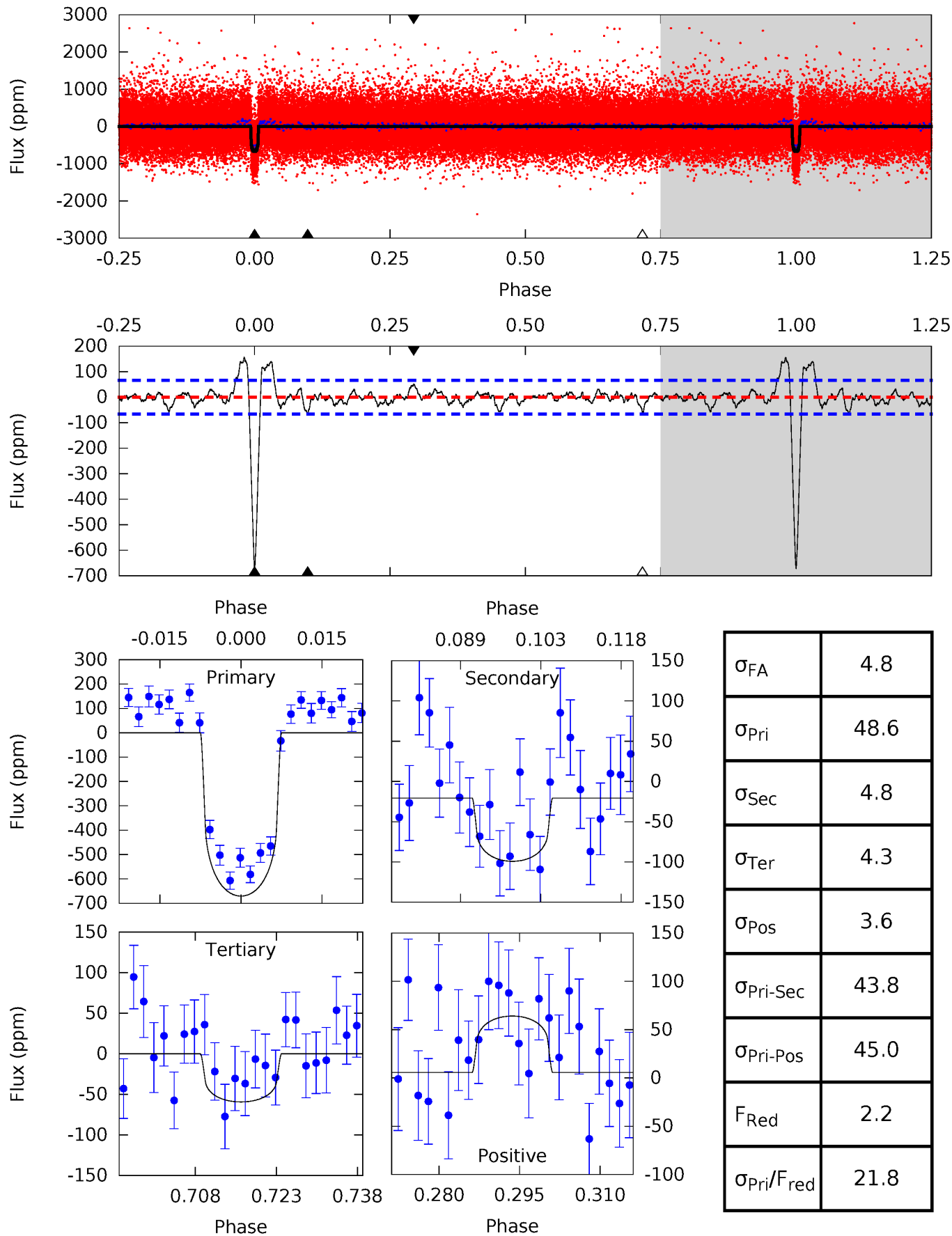
Flux Vetting - Quarter-Phased Transit Curves

TCE 004951877-01 P= 24.796290 Days T₀=145.5279 (BKJD)



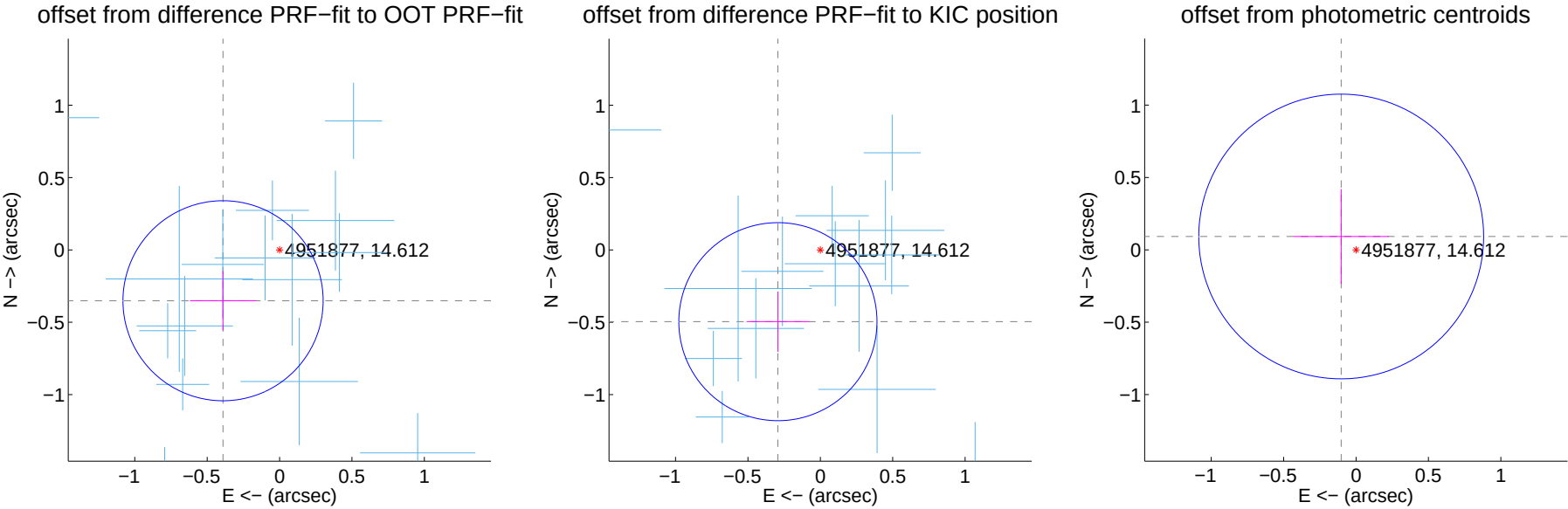
Flux Vetting - Model-Shift Uniqueness Test

TCE 004951877-01, P = 24.796290 Days



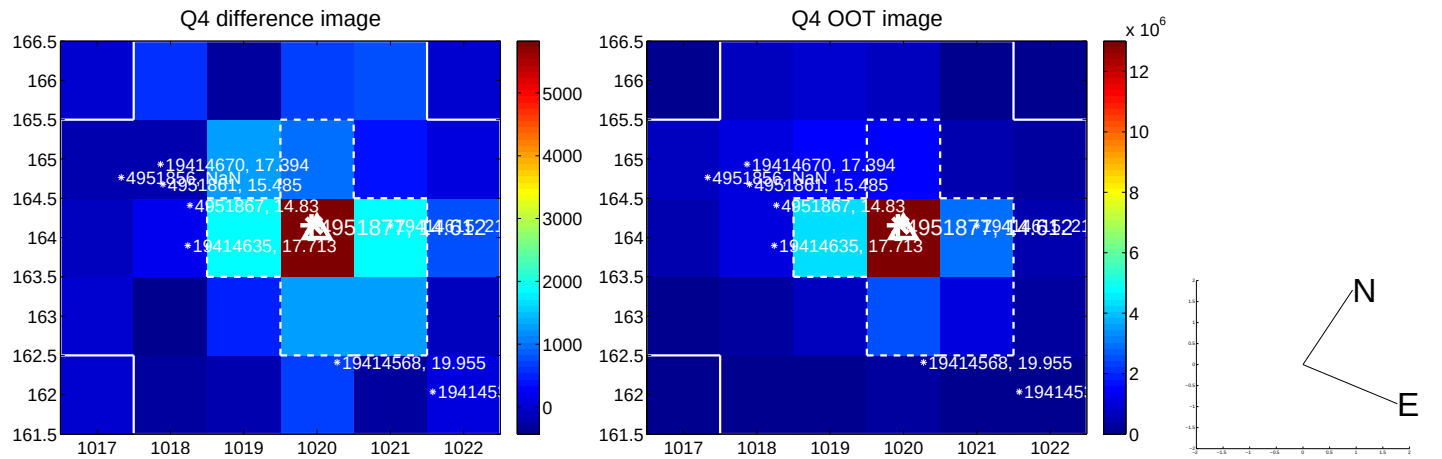
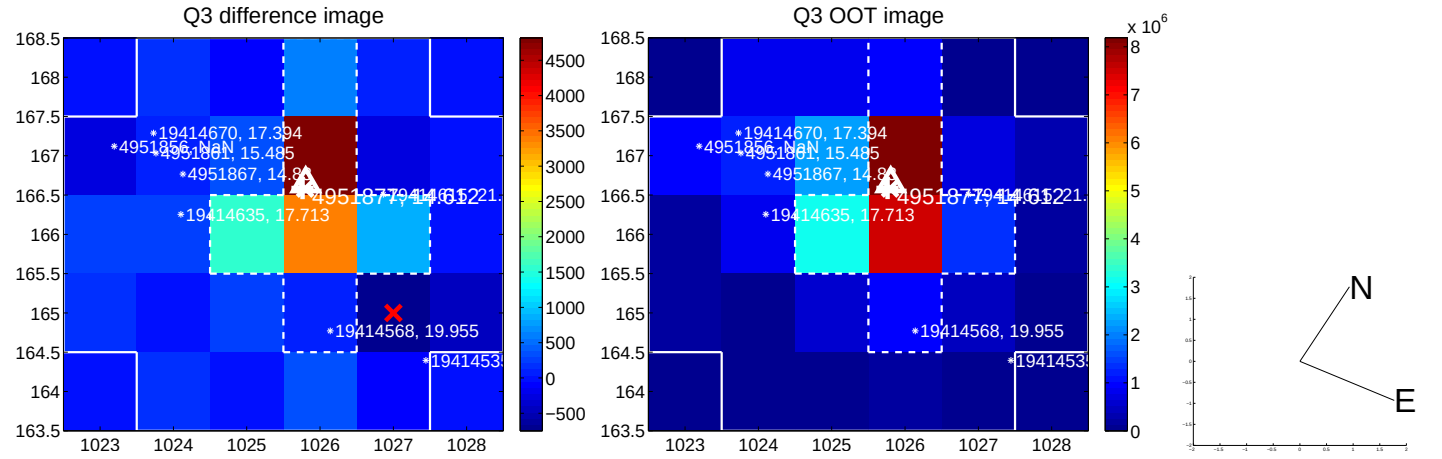
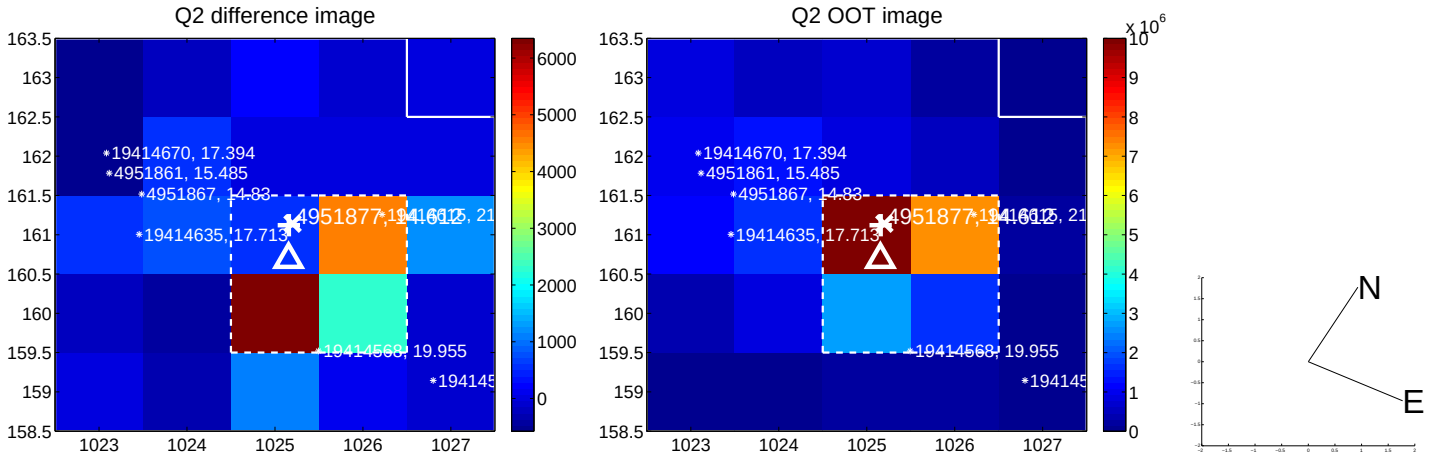
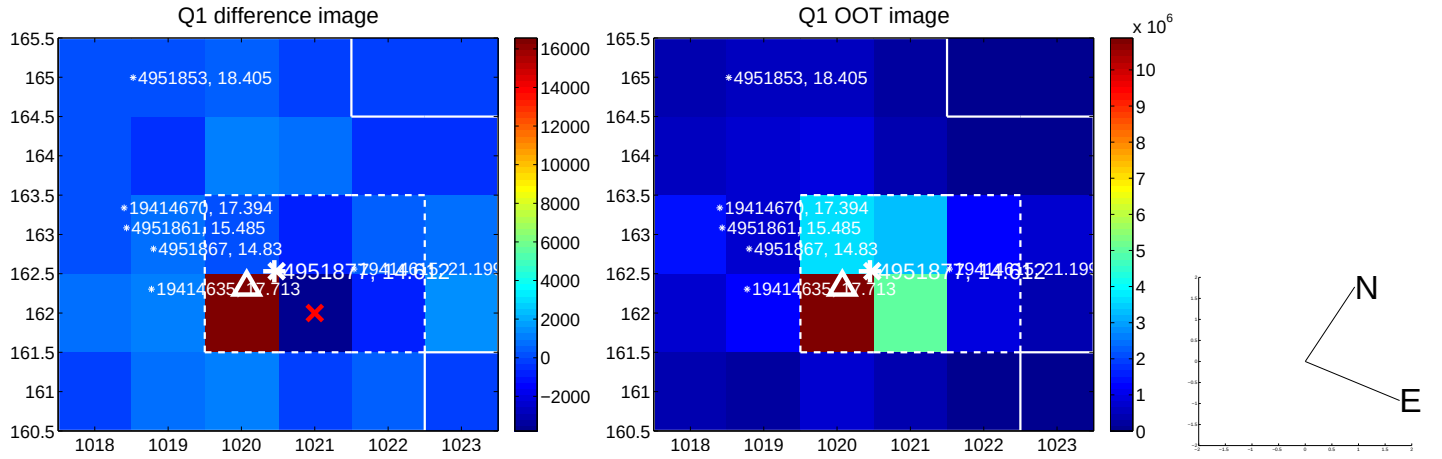
Supplemental centroid analysis for 004951877-01. Kepler magnitude: 14.61. Transit SNR 34.33
 There are 16 quarters with good PRF difference image offsets
 The direct PRF centroid is offset from the target star catalog position by about 0.14 arcsec

	Distance in arcsec	Distance / σ	Δ RA	Δ Dec
PRF-fit source offset from OOT	0.525 ± 0.230	2.28	0.390 ± 0.229	-0.351 ± 0.204
PRF-fit source offset from KIC position	0.576 ± 0.228	2.53	0.293 ± 0.216	-0.496 ± 0.209
photometric centroid source offset	0.14 ± 0.33	0.42	0.10 ± 0.33	0.09 ± 0.33

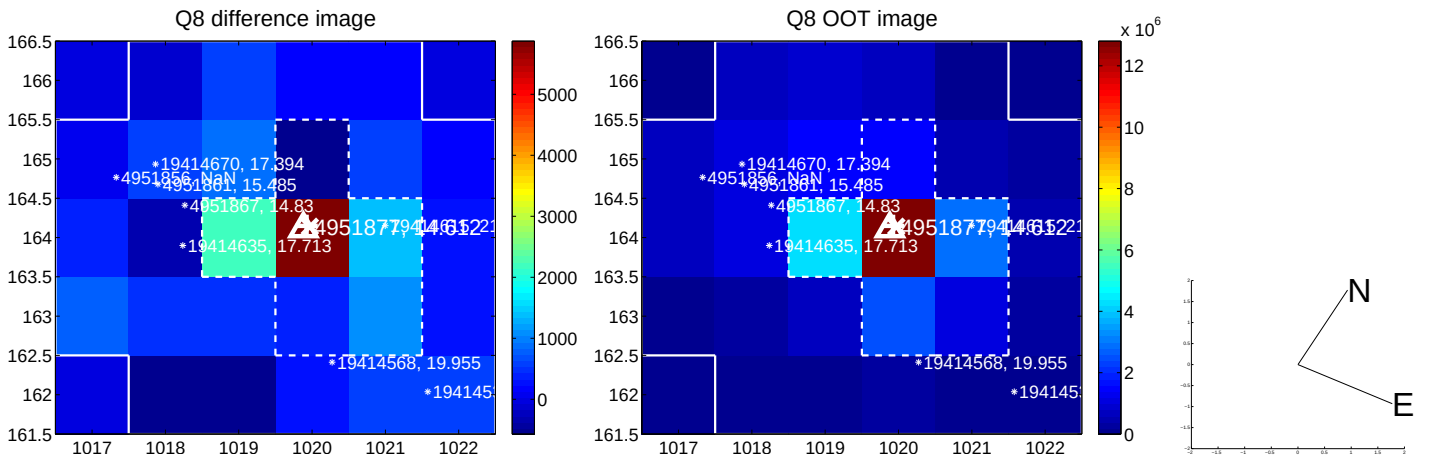
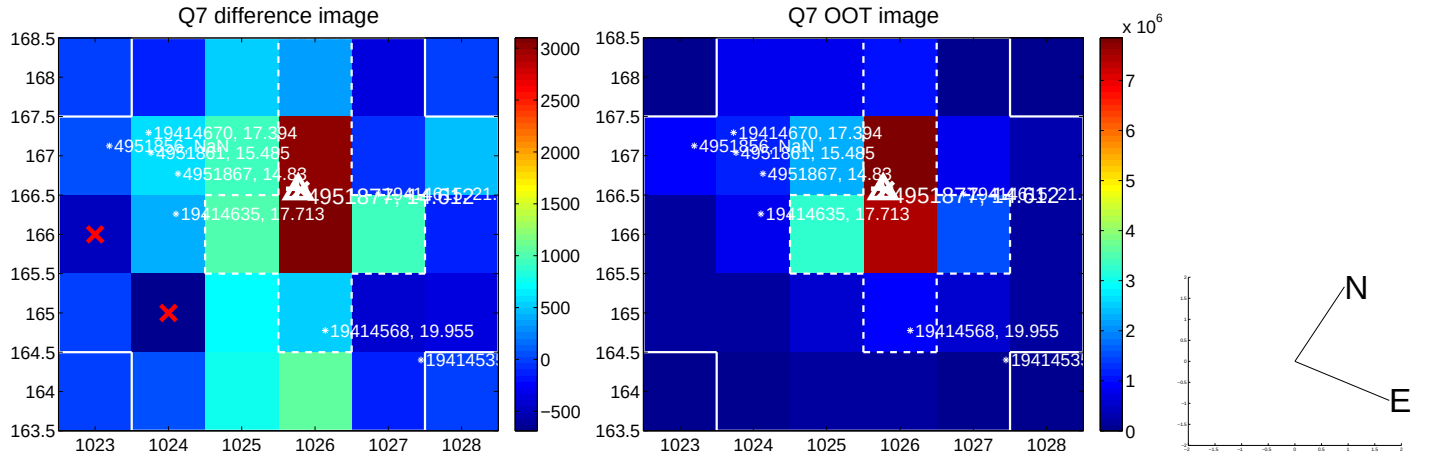
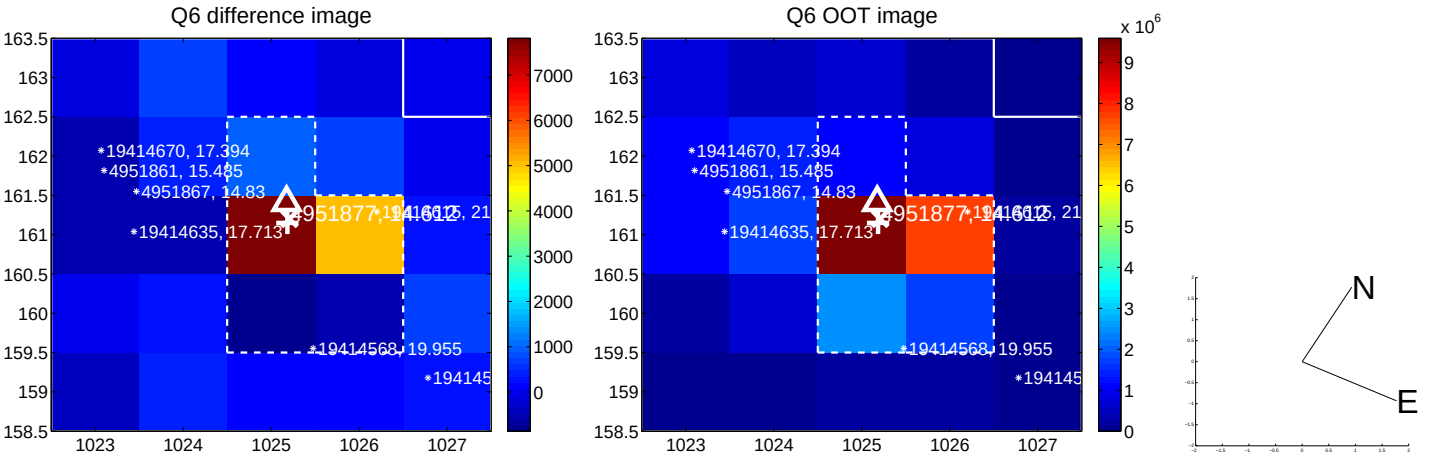
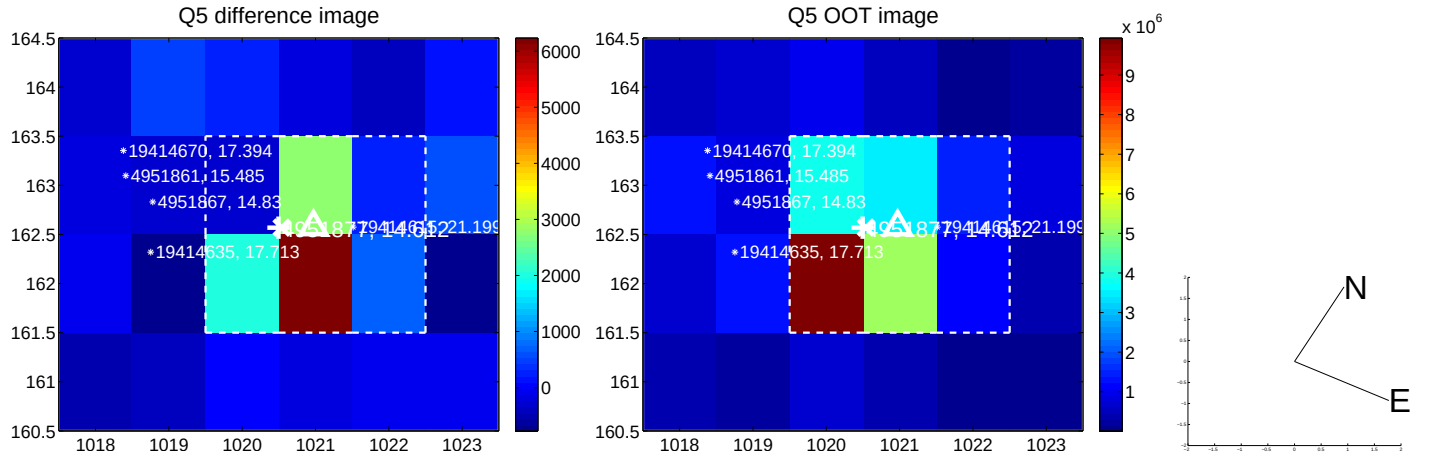


Centroid source offsets from the target star reconstructed from PRF and photometric centroids. Sky blue crosses: good quarterly centroid offsets; Vermillion crosses: bad quarterly centroid offsets; magenta cross: average over quarters. Length of the crosses: one- σ uncertainty. Blue circle: three- σ . Red *: target star. Blue *: Other stars. Text next to a star gives its KIC ID and kepmag. KIC IDs > 15,000,000 are from the UKIRT catalog.

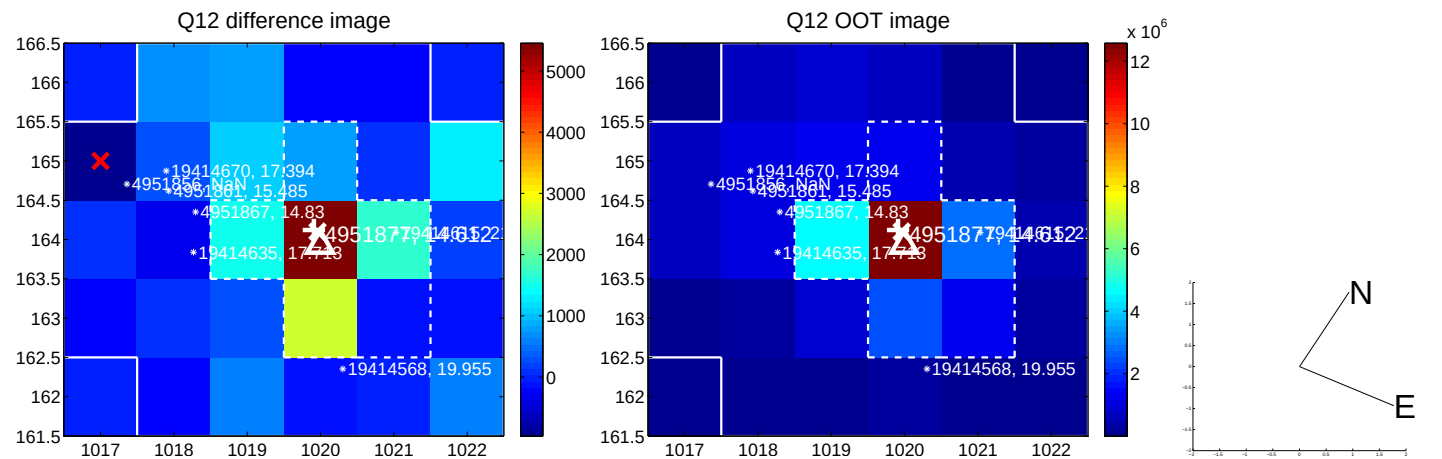
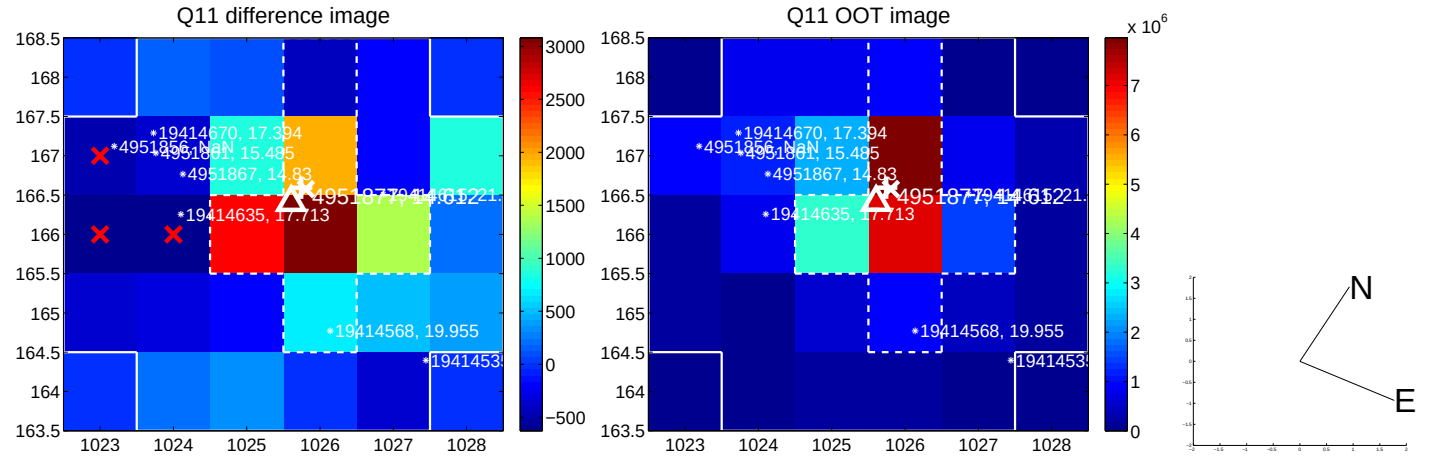
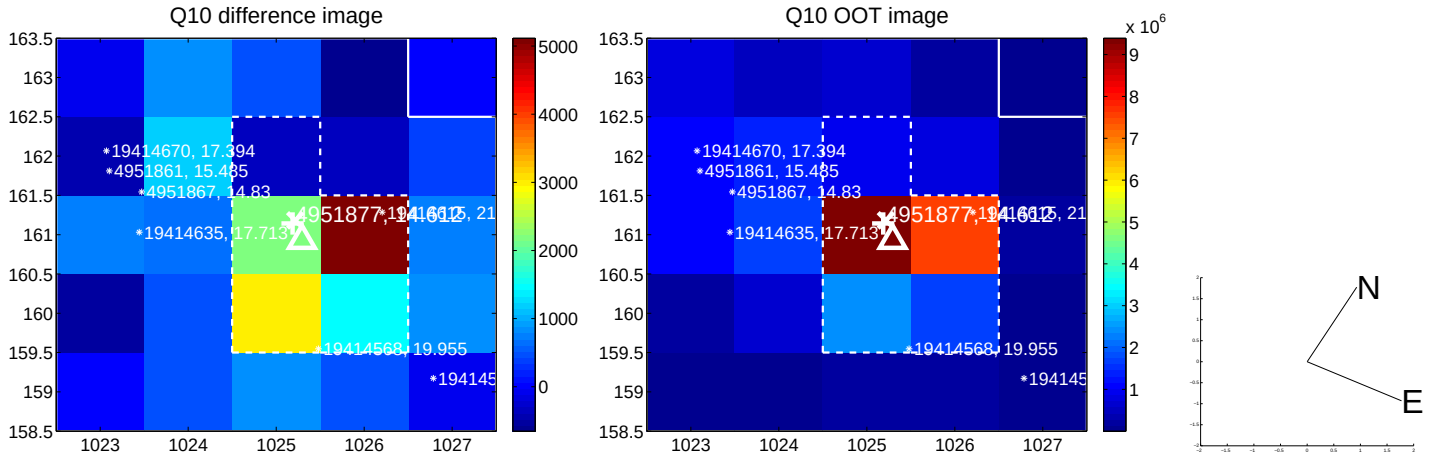
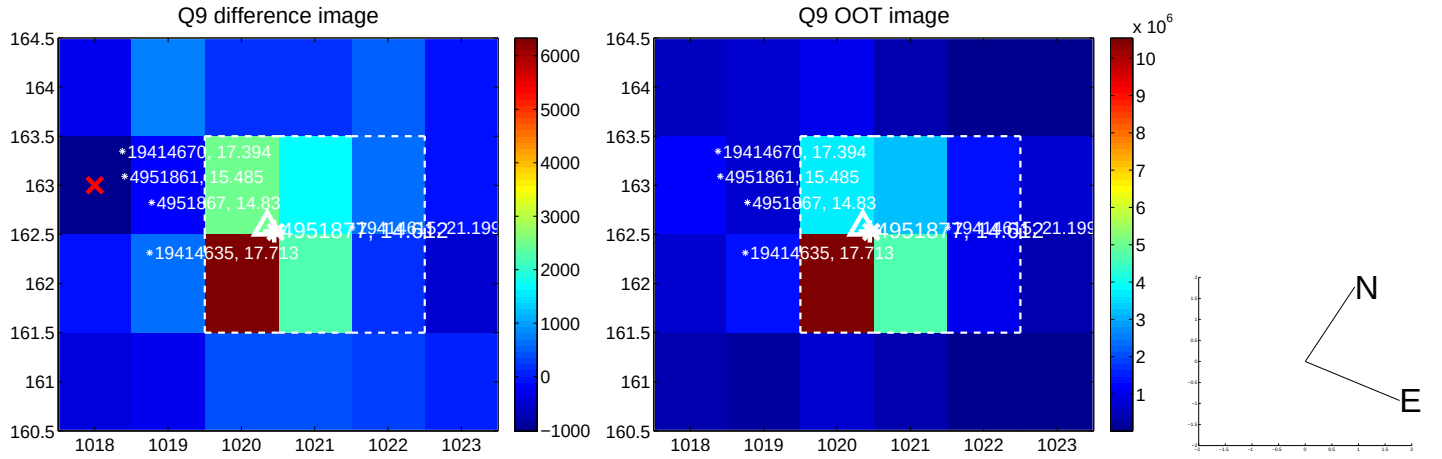
white $\times$: KIC target position; +: OOT centroid; $\triangle$: difference centroid. red $\times$: large negative pixel value.



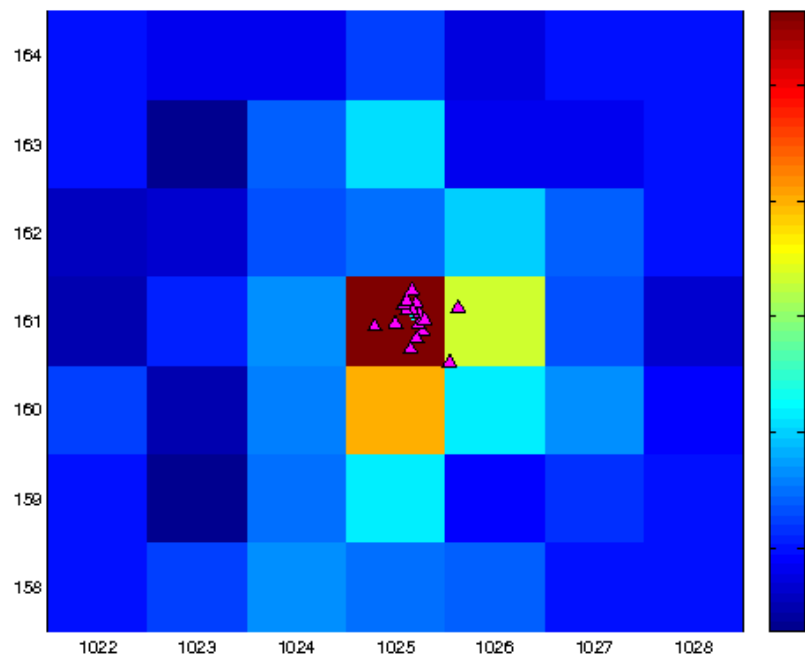
white $\times$: KIC target position; $+$: OOT centroid; $\triangle$: difference centroid. red $\times$: large negative pixel value.



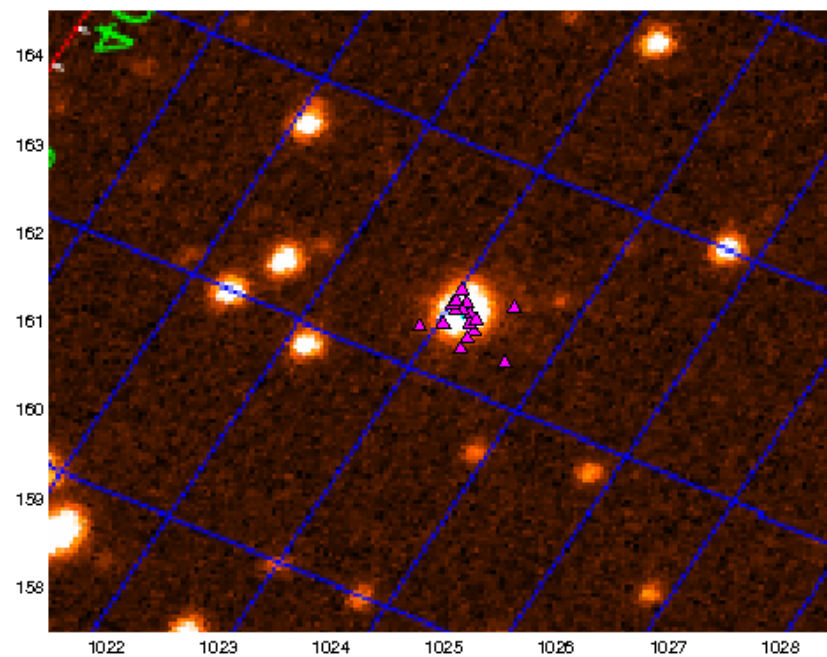
white $\times$: KIC target position; +: OOT centroid; $\triangle$: difference centroid. red $\times$: large negative pixel value.



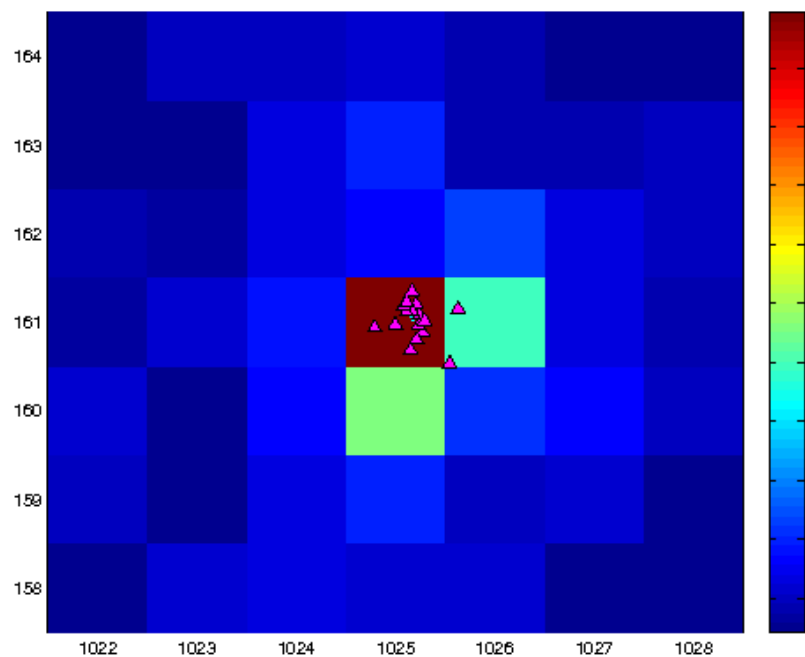
Average SNR During Transit



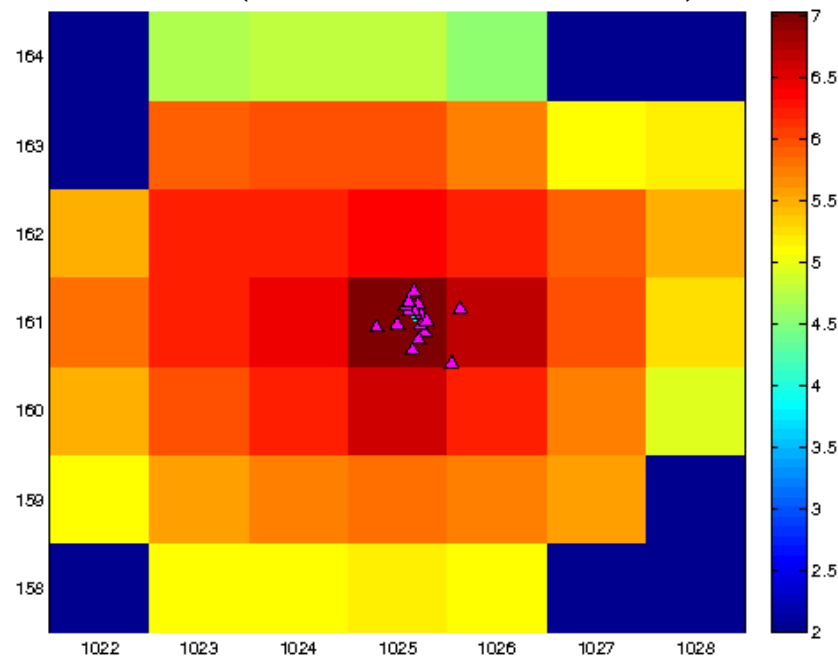
UKIRT Image



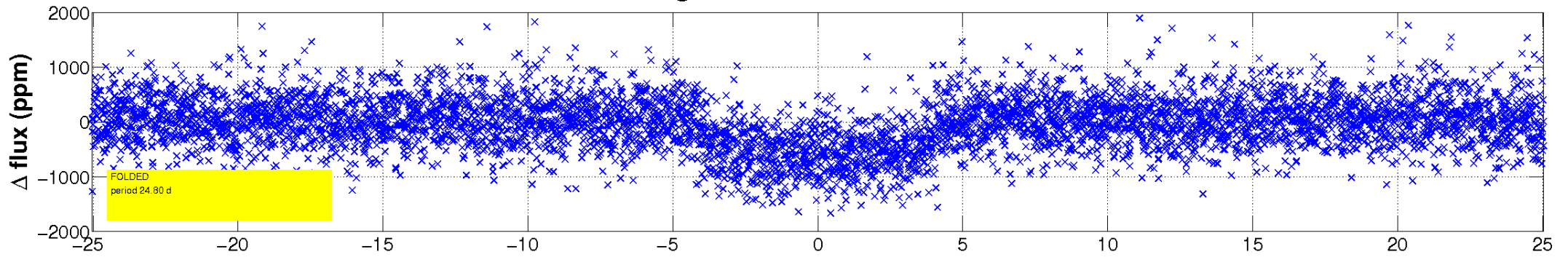
Average Depth During Transit



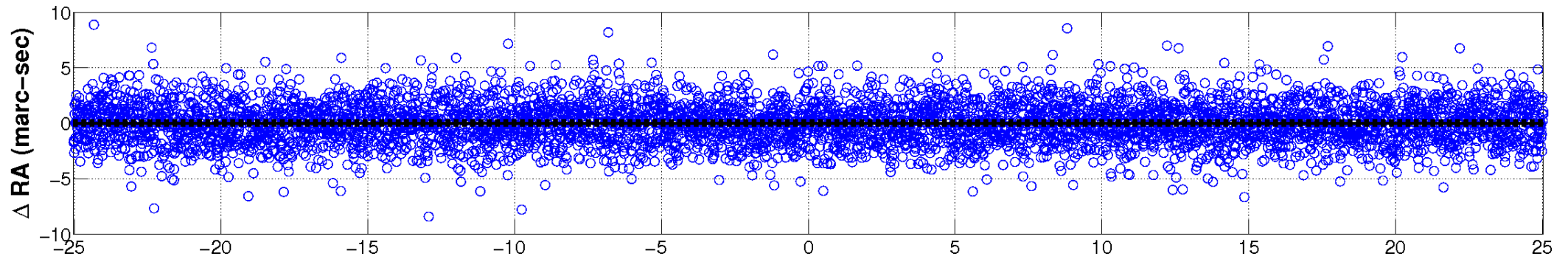
$\log_{10}(\text{Out-of-transit Flux})$



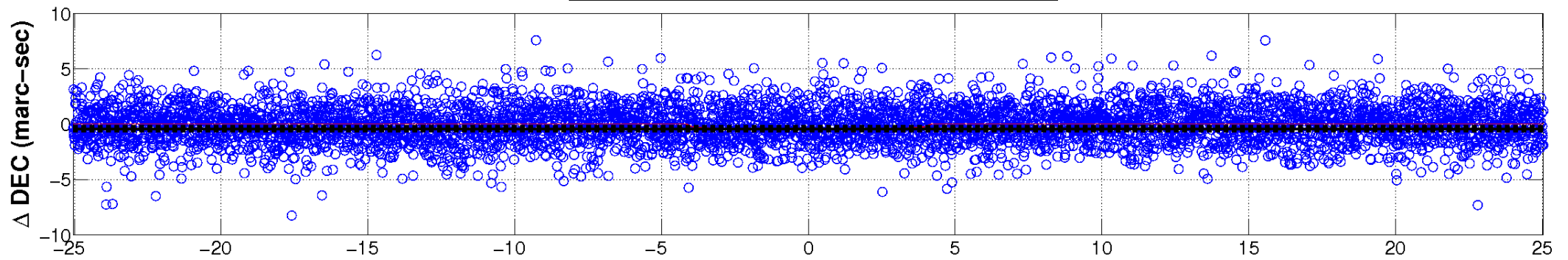
fluxWeightedCentroids, Planet 1 of 1



* detrended flux



○ detrended centroid — fit to transit model ➔ peak offset



Orbital Phase (hours)